

# Advanced Mathematical Determination of Optimal Smoothing Factors for Dynamic Signal Processing

## Introduction to the Smoothing Trade-Off and System Dynamics

In digital signal processing, modern control theory, and algorithmic time-series analysis, the mitigation of high-frequency noise without compromising the fidelity of underlying signal dynamics remains a paramount mathematical challenge. Sensors, financial markets, metrological instruments, and physical control interfaces generate data that is inherently corrupted by stochastic noise, multipath fading, micro-volatility, or systemic quantization errors. To extract the true underlying signal, practitioners rely on low-pass filters, most notably the Exponential Moving Average (EMA) or single-pole infinite impulse response (IIR) filter. The core operational parameter of an EMA is the smoothing factor, herein denoted as  $\gamma$  (frequently represented as  $\alpha$  in domain literature). This parameter determines the precise mathematical weight applied to the most recent observation relative to the historical state of the system. The selection of  $\gamma$  dictates a strict, inescapable mathematical trade-off between two opposing performance metrics: responsiveness and boundary stability. Responsiveness is defined as the minimization of phase delay, lag, or tracking error during signal transients. Boundary stability is defined as the minimization of jitter, variance, or output volatility during steady-state operations.

A high  $\gamma$  value approaches unity, heavily weighting new data. This configuration yields a highly responsive filter that tracks rapid signal changes with minimal phase delay, but fails to adequately attenuate high-frequency noise, resulting in degraded boundary stability. Conversely, a low  $\gamma$  value approaches zero, heavily weighting the historical state. This creates a highly stable output that effectively neutralizes jitter and stochastic variations, but introduces significant phase delay, rendering the system sluggish and unresponsive to genuine structural breaks in the data stream.

This report presents an exhaustive mathematical derivation of the optimal smoothing factor  $\gamma$ . It demonstrates that while a static optimal value can be calculated for strictly stationary environments by minimizing a composite risk function, real-world systems are rarely stationary. Consequently, achieving true optimization necessitates transitioning from a static parameter to dynamic, adaptive smoothing architectures. By analyzing cutting-edge methodologies—such as Kaufman's Adaptive Moving Average (KAMA) in financial econometrics, the 1-Euro Filter in human-computer interaction, and Model Predictive Control (MPC) in electromechanical systems—this analysis provides a rigorous framework for continuous, mathematically optimal parameter modulation. Furthermore, the implications of these filters are explored within the context of hardware implementation, energy storage systems, metrology, and advanced control topologies.

## Mathematical Formulations of the Exponential Moving

## Average

To determine the optimal value of the smoothing factor, the filter must first be rigorously defined in the discrete-time domain. The first-order exponential smoothing filter operates via a recursive difference equation:

$$Y_t = \gamma X_t + (1 - \gamma) Y_{t-1}$$

Where:

- $Y_t$  is the filtered output at the current discrete time step  $t$ .
- $X_t$  is the raw, noisy input observation sampled at time  $t$ .
- $\gamma \in [0, 1]$  is the dimensionless signal attenuation coefficient or smoothing factor.

## Relationship to Continuous-Time Systems and Time Constants

The discrete-time EMA is the digital equivalent of an analog single-pole low-pass filter, such as a traditional resistor-capacitor (RC) circuit. To understand the physical and mathematical meaning of  $\gamma$ , it is necessary to relate it to the continuous-time domain via the sampling period  $\Delta T$  and the continuous relaxation time constant  $\tau$ .

In a continuous-time dynamic system, the step response of a first-order low-pass filter is governed by the equation  $1 - e^{-t/\tau}$ . When mapped to a discrete sampled system, the equivalent smoothing factor can be derived exactly using the matched Z-transform or approximated via the backward Euler method. The exact analytical relationship linking these variables is given by:

$$\gamma = 1 - e^{-\frac{\Delta T}{\tau}}$$

For high sampling rates where the sampling period is significantly smaller than the time constant ( $\Delta T \ll \tau$ ), the Taylor series expansion of the exponential function allows for a widely utilized linear approximation:

$$\gamma \approx \frac{\Delta T}{\tau + \Delta T}$$

This relationship demonstrates that the smoothing factor is not an arbitrary dimensionless coefficient but rather a direct mathematical proxy for the system's time constant relative to its discrete sampling rate. A large  $\tau$  represents heavy filtering and dictates a small  $\gamma$ , while a small  $\tau$  represents light filtering and dictates a  $\gamma$  closer to 1. In frequency domain analysis, the time constant uniquely dictates the cutoff frequency  $f_c$  (the -3dB attenuation point) of the filter, where  $\tau = \frac{1}{2\pi f_c}$ . Therefore, any algorithm that modulates  $\gamma$  is mathematically equivalent to modulating the cutoff frequency of the physical filter.

## Analytical Optimization of the Static Smoothing Factor

To mathematically determine the absolute optimal value for  $\gamma$ , an objective cost function must be established. Let the observed signal  $X_t$  be composed of a true underlying deterministic signal  $S_t$  and an additive stochastic noise component  $V_t$ , such that  $X_t = S_t + V_t$ . Assume  $V_t$  is zero-mean white Gaussian noise with a known variance  $\sigma_v^2$ .

The mathematical objective is to minimize the Mean Squared Error (MSE) between the filtered output  $Y_t$  and the unobservable true signal  $S_t$ :

$$J(\gamma) = \mathbb{E}[(Y_t - S_t)^2]$$

This objective function can be decomposed into two distinct, inversely correlated components that perfectly represent the fundamental engineering trade-off:

1. **Variance (Boundary Stability):** The steady-state error caused by the transmission of ambient input noise through the filter architecture.
2. **Bias (Responsiveness):** The dynamic tracking error caused by the filter's inherent phase delay when attempting to follow changes in the true signal  $S_t$ .

## Component 1: Variance and Boundary Stability Analysis

When the true underlying signal is held constant ( $S_t = C$ ), the steady-state variance of the output  $Y_t$  is determined entirely by the noise power transfer characteristics of the filter. Taking the statistical variance of both sides of the EMA difference equation yields:

$$\sigma_y^2 = \gamma^2 \sigma_v^2 + (1 - \gamma)^2 \sigma_y^2$$

By algebraically isolating the output variance  $\sigma_y^2$ , the steady-state noise transmission equation is revealed:

$$\sigma_y^2 [1 - (1 - \gamma)^2] = \gamma^2 \sigma_v^2 \quad [2\gamma - \gamma^2] = \gamma^2 \sigma_v^2 \quad \sigma_y^2 = \frac{\gamma}{2 - \gamma} \sigma_v^2$$

This proof definitively establishes that as  $\gamma \rightarrow 0$ , the output variance  $\sigma_y^2 \rightarrow 0$ , thereby maximizing boundary stability and entirely suppressing noise. Conversely, as  $\gamma \rightarrow 1$ ,  $\sigma_y^2 \rightarrow \sigma_v^2$ , meaning the filter becomes perfectly transparent to high-frequency noise, offering zero boundary stability.

## Component 2: Bias and Dynamic Responsiveness

To evaluate the responsiveness of the system, one must analyze the filter's mathematical reaction to a dynamic, non-stationary signal. Assume the true signal is a linear ramp function,  $S_t = v \cdot t \cdot \Delta T$ , where  $v$  represents the constant rate of change (velocity or trend momentum).

If a ramp function is passed through a first-order low-pass filter, the output slope will eventually match the slope of the input, but the signal will be delayed by a constant temporal phase lag  $d$ . For the EMA architecture, the steady-state lag is equal to the time constant  $\tau$ . Expressing this temporal lag in terms of absolute signal amplitude error yields:

$$\text{Error}_{\text{lag}} = v \cdot \tau = v \cdot \frac{\Delta T (1 - \gamma)}{\gamma}$$

In this dynamic scenario, as  $\gamma \rightarrow 1$ , the lag error approaches zero, maximizing the filter's responsiveness. As  $\gamma \rightarrow 0$ , the lag error approaches infinity, completely decoupling the output trajectory from the true signal.

## Solving the Optimization Polynomial

Combining the variance and bias components yields the composite risk function representing the total Mean Squared Error of the filter:

$$J(\gamma) = \left( v \cdot \Delta T \cdot \frac{1 - \gamma}{\gamma} \right)^2 + \frac{\gamma}{2 - \gamma} \sigma_v^2$$

To find the optimal static smoothing factor, denoted as  $\gamma^*$ , the first derivative of the cost function with respect to  $\gamma$  must be computed and set to zero:

$$\frac{dJ}{d\gamma} = 0$$

Taking the derivative:

$$\frac{d}{d\gamma} \left[ v^2 \Delta T^2 \left( \frac{1 - \gamma}{\gamma} \right)^2 \right] + \frac{d}{d\gamma} \left[ \sigma_v^2 \left( \frac{\gamma}{2 - \gamma} \right) \right] = 0 \quad -2 v^2 \Delta T^2 \frac{1 - \gamma}{\gamma^3} + \sigma_v^2 \frac{2}{(2 - \gamma)^2} = 0$$

Solving this non-linear polynomial for  $\gamma$  yields the theoretically optimal static smoothing factor. However, this exact mathematical derivation reveals a critical dependency: the optimal value of  $\gamma$  is inextricably linked to both the ambient noise variance  $\sigma_v^2$  and the signal velocity  $v$ . If the operational environment is strictly stationary—meaning the noise profile and the rate of change of the signal never vary—a single, static  $\gamma^*$  can be hard-coded into the microcontroller or trading algorithm to achieve perpetual optimal performance.

## Metrology and Noise Profiling: Allan Deviation (ADEV) and Time Constants

The pursuit of this stationary optimum is highly visible in precision metrology, particularly in the post-processing of data from high-resolution Digital Multimeters (DMMs). When measuring stable voltage references, metrologists utilize statistics such as the Allan Deviation (ADEV), denoted as  $\sigma(\tau)$ , to determine the mathematically ideal integration time constant for a given instrument and environment.

Allan Deviation provides a graphical and mathematical representation of frequency stability and noise over various observation times. In a standard scenario involving a 34401A DMM operating with Auto-Zero enabled, a setting of 100 NPLC (Number of Power Line Cycles) results in a fixed discrete sampling period of approximately 4.2 seconds.

By plotting the ADEV  $\sigma(\tau)$  against various integration times, practitioners identify a global minimum—the precise point where high-frequency white noise has been sufficiently averaged out, but before low-frequency  $1/f$  flicker noise or thermal drift begins to corrupt the measurement. If the ADEV minimum indicates an optimal integration time  $\tau = 40$  seconds, the metrologist must construct a post-processing pipeline that aligns with this target.

Achieving this 40-second time constant can involve a combination of running medians (to eliminate non-Gaussian outliers) and an EMA low-pass filter. The relationship derived earlier ( $\tau = \frac{\Delta T (1 - \gamma)}{\gamma}$ ) allows the exact required  $\gamma$  to be calculated. With  $\Delta T = 4.2$  seconds and a target  $\tau = 40$  seconds, the optimal  $\gamma$  is approximately 0.095. This value perfectly balances the attenuation of ambient electromagnetic interference against the tracking of legitimate long-term voltage drift. However, this optimal value is only valid because the metrological environment (a highly controlled laboratory voltage reference) is fundamentally stationary.

## The Non-Stationary Reality: Necessity of Adaptive Filtering Topologies

In virtually all practical applications outside of controlled laboratory metrology—ranging from financial asset pricing to physical robotic tracking, and energy grid load balancing—the assumption of stationarity is fatally flawed. The underlying velocity of the signal  $v$  is highly dynamic. Financial markets undergo periods of sideways ranging (low  $v$ ) interspersed with violent trend breakouts (high  $v$ ). Similarly, a human operator interacting with a digital interface will hold a stylus perfectly still (introducing pure jitter at  $v=0$ ) and then execute rapid sweeping motions (introducing high lag penalty at large  $v$ ).

Because the mathematically optimal  $\gamma$  strictly depends upon  $v$ , and  $v$  is actually a time-varying parameter  $v_t$ , it is mathematically impossible for any static  $\gamma$  to be optimal across all operational regimes. This limitation necessitates the use of non-linear, adaptive filters

where  $\gamma$  becomes a dynamically calculated variable,  $\gamma_t$ , updated at each discrete sampling interval based on the localized characteristics of the data sequence.

Adaptive smoothing topologies continuously solve the optimization polynomial in real-time. By continuously estimating the signal velocity and noise characteristics, these algorithms expand or contract the filter's bandwidth (cutoff frequency) on the fly. Two of the most robust implementations of this principle are Kaufman's Adaptive Moving Average (KAMA) in econometrics and the 1-Euro Filter in spatial kinematics.

## Kaufman's Adaptive Moving Average (KAMA) in Econometrics

Introduced by quantitative analyst Perry J. Kaufman in 1995, KAMA was explicitly engineered to address the lag-stability trade-off in financial time-series data. Traditional moving averages frequently subject algorithmic trading models to catastrophic "whipsaw" losses during volatile, trendless markets due to poor boundary stability, while simultaneously missing optimal entry points during strong breakouts due to inherent phase delay.

KAMA resolves this paradox by defining an adaptive smoothing factor based on a fractal efficiency metric known as the Efficiency Ratio (ER).

### The Efficiency Ratio (ER) Calculation

The Efficiency Ratio is a mathematical quantification of the signal's directional persistence (velocity) relative to its total path length (volatility) over a rolling lookback period of  $n$  samples. It isolates the underlying trend from ambient market noise.

The ER is calculated via two intermediate components:

1. **Direction (Momentum):** The absolute net change in the asset's closing price over  $n$  periods.  $\text{Direction}_t = |X_t - X_{t-n}|$
2. **Volatility (Path Length):** The sum of the absolute bar-by-bar price variations over the exact same  $n$  periods.  $\text{Volatility}_t = \sum_{i=0}^{n-1} |X_{t-i} - X_{t-i-1}|$

The Efficiency Ratio is then the quotient of these two values:

$$ER_t = \frac{\text{Direction}_t}{\text{Volatility}_t}$$

By definition,  $ER_t \in [0, 1]$ .

- If the price moves monotonically in a single direction for  $n$  periods, the net change equals the absolute sum of individual changes, yielding an ER of 1. This indicates maximum efficiency, zero noise, and a strong directional trend.
- If the price oscillates wildly but ends at the exact same value it started  $n$  periods prior, the net change is zero while the sum of individual changes is massive, yielding an ER of 0. This indicates pure chop, pure noise, and zero structural velocity.

### Deriving the Smoothing Constant (SC) and the Squaring Mechanism

KAMA translates the ER into an adaptive smoothing factor (referred to in quantitative finance literature as the Smoothing Constant, or SC). The mechanism defines a bounding range for the filter's responsiveness, utilizing a "Fastest SC" and a "Slowest SC" limit.

Standard KAMA implementations (such as those in the TA-Lib library) default to the following bounds, though they are mathematically tunable:

- **Fastest EMA Length:** 2 periods, yielding  $\text{FastSC} = \frac{2}{2+1} \approx 0.6667$

- **Slowest EMA Length:** 30 periods, yielding  $\text{SlowSC} = \frac{1}{2^{30+1}} \approx 0.0645$

At each time step, an intermediate scaling value is computed via linear interpolation based on the current ER:

$$\text{Scaled\_SC} = \text{ER}_t \times (\text{FastSC} - \text{SlowSC}) + \text{SlowSC}$$

Crucially, Kaufman introduced a non-linear transformation to this scaled value to heavily penalize noise. The final dynamically calculated smoothing factor is the square of the interpolated value:

$$\gamma_t = \text{SC}_t = \left[ \text{ER}_t \times (\text{FastSC} - \text{SlowSC}) + \text{SlowSC} \right]^2$$

The mathematical decision to square the interpolation term is the precise mechanism that enforces boundary stability. Squaring a fractional value between 0 and 1 disproportionately reduces smaller values while having a much less pronounced effect on values approaching 1.

| Market Condition          | Efficiency Ratio (ER) | Unsquarred Interpolation | Final $\gamma_t$ (Squared) | Algorithmic Behavior                                   |
|---------------------------|-----------------------|--------------------------|----------------------------|--------------------------------------------------------|
| <b>Sideways / Choppy</b>  | $\approx 0.00$        | 0.0645                   | <b>0.0041</b>              | Maximum boundary stability; flatlines to ignore noise. |
| <b>Weak Trend / Mixed</b> | $\approx 0.30$        | 0.2452                   | <b>0.0601</b>              | Heavy smoothing; highly cautious tracking.             |
| <b>Moderate Trend</b>     | $\approx 0.60$        | 0.4258                   | <b>0.1813</b>              | Balanced responsiveness and smoothing.                 |
| <b>Strong Breakout</b>    | $\approx 1.00$        | 0.6667                   | <b>0.4444</b>              | Maximum responsiveness; minimal lag penalty.           |

When the market is highly inefficient ( $\text{ER} \approx 0$ ), squaring reduces the final  $\gamma_t$  to approximately 0.004. This extremely low value creates a filter with an immense time constant, effectively flatlining the moving average and rendering it entirely immune to the ambient market jitter. Conversely, during a breakout ( $\text{ER} \approx 1$ ), squaring yields a  $\gamma_t$  of approximately 0.444. While slightly dampened, this remains highly aggressive, allowing the filter to tightly track the rapid price trajectory with minimal phase delay.

## Strategic Integration and Position Sizing

In deployment, quantitative traders use KAMA not just for smoothing, but as a core component of trading logic. KAMA acts as a filter to screen out choppy markets where breakouts are frequently false positives ("fakeouts"). It determines whether to apply a trend-following algorithm (when ER is high) or a mean-reversion algorithm (when ER is low).

Furthermore, KAMA signals are frequently corroborated using momentum oscillators. A long entry opportunity is validated when the asset price sits above the KAMA line (indicating a

structural uptrend), but the Relative Strength Index (RSI) briefly dips below 30, signaling an oversold pullback within that efficient trend. Similarly, MACD crossovers carry significantly higher statistical weight when they align with the directional slope of the KAMA. Position sizing is dynamically modulated alongside the filter: capital allocation is increased when the ER is high, and drastically reduced when the ER drops, mathematically limiting portfolio risk during structurally unsound market regimes.

## Kinematic Tracking and the 1-Euro Filter

While KAMA utilizes fractal path length to gauge noise, applications in physical human-computer interaction (HCI), robotics, and augmented reality tracking face a different manifestation of the same optimization problem. In cursor tracking or 3D human spatial body tracking (e.g., enhanced Mediapipe pose detection frameworks), boundary stability is characterized by high-frequency spatial jitter when the user is attempting to hold perfectly still, whereas responsiveness is characterized by a trailing spatial lag when the user executes rapid motions.

The 1-Euro Filter, developed by Casiez, Roussel, and Vogel, solves this by directly modulating the cutoff frequency of a first-order low-pass filter using the instantaneous rate of change (discrete derivative) of the spatial signal.

### The Cutoff Frequency Modulation Algorithm

The 1-Euro Filter operates on the psychophysical premise that human perception of lag is highly acute during high-speed movements, whereas perception of jitter is acute during low-speed, precise movements.

The algorithm first computes the instantaneous spatial velocity of the noisy signal:

$$\hat{v}_t = \frac{X_t - X_{t-1}}{\Delta T}$$

Because the raw velocity  $\hat{v}_t$  is calculated by differentiating noisy position data, the velocity signal is itself highly contaminated. Therefore, the algorithm applies a static low-pass filter to the velocity signal with a constant cutoff frequency  $f_v$  to produce a smoothed velocity magnitude  $\bar{v}_t$ .

The core mathematical innovation is the linear mapping of this smoothed velocity magnitude directly to the cutoff frequency of the primary position filter:

$$f_{c,t} = f_{\min} + \beta \bar{v}_t$$

Where:

- $f_{\min}$  is the minimum cutoff frequency (in Hz), which dictates the baseline boundary stability of the filter when the signal is perfectly still ( $v=0$ ). A properly tuned lower  $f_{\min}$  entirely eliminates slow-speed jitter.
- $\beta$  is the adaptive speed coefficient. It determines how aggressively the cutoff frequency expands as structural velocity increases. A higher  $\beta$  rapidly eliminates phase delay during high-speed movements.

Once the dynamic cutoff frequency  $f_{c,t}$  is determined, it is converted into the discrete smoothing factor  $\gamma_t$  (often denoted  $\alpha$  in 1-Euro literature) using the standard Z-domain transformation:

$$\gamma_t = \frac{2\pi f_{c,t} \Delta T}{1 + 2\pi f_{c,t} \Delta T}$$

The final estimated position is then calculated using the standard EMA difference equation:

$$Y_t = \gamma_t X_t + (1 - \gamma_t) Y_{t-1}$$

## Two-Stage Empirical Optimization

The 1-Euro Filter represents a highly optimized, domain-specific solution to the composite  $J(\gamma)$  cost function. By isolating the two variables of interest (lag and jitter) into two entirely decoupled tuning parameters ( $f_{\min}$  and  $\beta$ ), it allows tracking engineers to empirically tune the system boundaries with extreme precision without causing systemic cross-contamination. To mathematically optimize a 1-Euro filter for a new tracking application, the tuning sequence is deterministic:

1. Set  $\beta = 0$  (rendering the filter entirely static) and hold the input target perfectly steady. Decrease  $f_{\min}$  until boundary stability is achieved and spatial jitter falls below the threshold of human perception or mechanical tracking tolerance.
2. Induce high-velocity, dynamic input signals. Gradually increase the coefficient  $\beta$  until the phase delay (lag) is eliminated, bringing dynamic responsiveness up to the required specification.

This two-step process mathematically guarantees an optimal parameter set for any physical kinematic tracking system, circumventing the need for computationally intensive Kalman filters in resource-constrained environments like embedded virtual reality headsets.

## Industrial Power Systems: PV+BESS Ramp-Rate Control Integration

The theoretical properties of smoothing factors extend significantly beyond software algorithms and visual graphical charts. In physical hardware and high-voltage power electronics, the parameter  $\gamma$  exerts direct, tangible effects on energy efficiency, thermal dynamics, and systemic grid stability.

A prominent industrial application of low-pass filtering occurs in renewable energy integration. Photovoltaic (PV) power generation is inherently volatile due to transient atmospheric conditions and moving cloud cover, leading to rapid fluctuations in output power that can violently destabilize local microgrid frequencies. To meet strict utility grid code requirements, PV plants are paired with Battery Energy Storage Systems (BESS) governed by continuous Ramp-Rate (RR) control algorithms.

The objective of RR control is to ensure that the active power injected into the grid,  $P_g$ , does not change by more than a mathematically specified percentage per minute (e.g., maximum 10%/min variance relative to rated capacity). Researchers frequently utilize EMA filters to smooth the raw PV power output  $P_{\text{pv}}$ , where the smoothed signal directly dictates the grid injection target  $P_g$ .

The choice of the smoothing factor  $\alpha$  (equivalent to  $\gamma$ ) directly impacts the battery's thermodynamic State of Charge (SOC). Because energy cannot be destroyed, the battery inverter must dynamically absorb or inject the exact power difference between the raw solar generation and the smoothed grid output:  $P_{\text{bess}} = P_{\text{pv}} - P_g$ .

If  $\gamma$  is configured too high (e.g.,  $>0.07$  on a 20-minute window), the filter is overly responsive. The grid output closely tracks the volatile solar output, and the ramp-rate mathematically exceeds the 10%/min legal threshold, resulting in systemic grid compliance failure and potential disconnection penalties.

If  $\gamma$  is configured too low (e.g., 0.04), the filter enforces strict boundary stability. The grid output becomes incredibly smooth, easily satisfying the 10%/min limitation. However, because

$P_g$  is artificially slow to react to massive surges in  $P_{pv}$ , the BESS must rapidly absorb the excess generation. Research evaluating EMA smoothing over a 20-minute window has demonstrated that utilizing an  $\alpha = 0.04$  forces the battery to absorb so much continuous energy that it reaches 100% SOC saturation. Once the storage medium is fully saturated, the control loop structurally fails; it can no longer absorb excess power, forcing raw fluctuations to bypass the system entirely and dump directly into the grid. Furthermore, such aggressive filtering prevents the battery from returning to its original state of charge at the end of the daily cycle, severely impacting the long-term economic viability of the plant.

| EMA Smoothing Factor ( $\alpha$ ) | RR Grid Compliance                   | Impact on Battery SOC                     | System Viability                                    |
|-----------------------------------|--------------------------------------|-------------------------------------------|-----------------------------------------------------|
| $\alpha \leq 0.04$                | Fully Compliant ( $<10\%/min$ )      | <b>Critical Failure</b> (100% Saturation) | Fails to inject generated power; clips arrays.      |
| $\alpha \approx 0.05$             | Compliant                            | High Cycling Wear                         | Borderline viability; requires massive BESS sizing. |
| $\alpha \geq 0.07$                | <b>Non-Compliant</b> ( $>10\%/min$ ) | Minimal Cycling Wear                      | Fails grid code regulations; utility penalties.     |

This physical dynamic represents the precise manifestation of the mathematical bias/variance trade-off mapped into thermodynamics. To solve this, advanced PV+BESS control architectures must implement *adaptive* smoothing factors, dynamically modulating  $\gamma$  based on the real-time derivative of the irradiance (to calculate exact RR compliance methods, such as the difference between two endpoints of a 60-second interval or maximum interval deviations) and the instantaneous available headroom within the BESS SOC parameters.

## Flowmeter Signal Processing and Programmable Logic Controllers (PLCs)

The deployment of mathematical filtering algorithms in embedded industrial environments requires specific adaptations to accommodate strict computational constraints and discrete signal sources. A fundamental example is the calculation of fluid flow rates using turbine or paddle-wheel flowmeters connected to Programmable Logic Controllers (PLCs).

Flowmeters generate discrete electrical pulses proportional to the volume of fluid passing through them (defined by a K-factor, e.g., pulses per gallon). The PLC utilizes a High-Speed Counter (HSC) to continuously track the lifetime accumulation of these pulses. To determine the instantaneous flow rate in Gallons Per Minute (GPM), the PLC must sample the accumulator on a fixed periodic interrupt and compute the delta over a known interval.

The core flow rate calculation is an algebraic discrete derivative:

$$\text{Flow Rate (GPM)} = \frac{(N - N_{\text{prev}}) \times 60}{K \times T}$$

Where  $N$  is the current pulse count,  $N_{\text{prev}}$  is the count at the previous sample,  $K$  is the K-factor, and  $T$  is the sampling period in seconds.

Because fluid dynamics are subject to turbulent flow noise, cavitation, and pump pulsation, the raw calculated GPM signal will exhibit severe high-frequency jitter. Industrial control systems route this raw GPM value through an EMA filter to create a stable process variable suitable for PID control loops.

However, lower-tier PLCs and microcontrollers lack dedicated Floating-Point Units (FPUs). Executing floating-point multiplication in software consumes significant clock cycles, which can cause jitter in the execution time of real-time control loops. A highly efficient mathematical workaround involves restricting the smoothing factor to inverse powers of two:

$$\gamma = \frac{1}{2^k}$$

By defining  $\gamma$  in this manner, the computationally expensive floating-point multiplication required by the EMA equation can be entirely replaced by a highly efficient right bit-shift operation.

$$EMA_n = EMA_{n-1} + [(X_n - EMA_{n-1}) \gg k]$$

In this implementation, the integer parameter  $k$  replaces  $\gamma$  as the primary tuning variable. A larger  $k$  corresponds to an exponentially smaller  $\gamma$ , shifting the pole of the transfer function closer to  $z=1$  on the unit circle, thereby drastically lowering the cutoff frequency and increasing boundary stability. While this optimization limits the resolution of  $\gamma$  to discrete fractional steps (0.5, 0.25, 0.125), it provides a sufficiently granular mechanism for empirical PLC tuning while accelerating computational throughput by orders of magnitude.

## Embedded Control Systems: MPC, ADRC, and Vondrak Filters

Adaptive filtering serves as the foundational data conditioning layer for highly advanced, predictive control architectures. When dealing with complex electromechanical systems—such as Permanent Magnet Synchronous Motor (PMSM) drives used in aircraft electric propulsion—traditional Proportional-Integral (PI) controllers suffer from unavoidable phase lag during rapid disturbance variations.

Modern aerospace architectures utilize Artificial Neural Network-based Model Predictive Control (ANN-MPCC) or Adaptive Network-based Fuzzy Inference Systems (ANFIS) combined with Active Disturbance Rejection Control (ADRC). These systems require impeccably clean derivative data to reconstruct real-time disturbance rates of change for feedforward compensation.

To achieve this without compromising prediction accuracy, controllers utilize orthogonal Laguerre functions combined with an adaptive smoothing factor. By leveraging the orthogonal symmetry of the Laguerre basis functions and the structural invariance of continuous-time speed dynamics, the adaptive smoothing factor reduces computational complexity while heavily suppressing the chattering phenomenon inherent in sliding mode control surfaces.

The empirical results of integrating adaptive smoothing into predictive models are stark. In comparative tests, traditional PI-MPCC approaches exhibit significant systemic overshoot and harmonic distortion. Conversely, the ANFIS-MPCC utilizing dynamic parameter adaptation reduces control overshoot to a mere 0.015%, minimizes settling time to 0.00147 seconds, and heavily suppresses Total Harmonic Distortion (THD) down to 2.0% at rated speeds.

Furthermore, discrete-time Lyapunov stability analysis is used to rigorously prove that the estimation errors of the internal system states under these adaptively filtered schemes remain Uniformly Ultimately Bounded (UUB).

In parallel domains, such as the initial alignment of Inertial Measurement Unit (IMU) sensors on disturbed bases (e.g., vehicles subjected to wind buffeting), Vondrak low-pass filters are implemented. A Vondrak filter utilizes a genetic algorithm to optimally select its smoothing factor based on spectral analysis of the force observation. This method decisively eliminates high-frequency mechanical noise without requiring complete mathematical modeling of the

system's angular motions, proving vastly superior to standard Adaptive Kalman Filters (AKF) in non-linear disturbed environments.

## **Physical Hysteresis, Electromechanical Wear, and Actuator Stability**

In physical electromechanical domains, a lack of boundary stability in signal processing translates directly into mechanical hysteresis, severe physical wear, and acoustic noise. Academic research generated by institutions such as Oregon State University and Portland State University has extensively analyzed the intersection of adaptive filtering and physical closed-loop control.

Consider the operation of a strain gage transducer system used to measure structural load. The gage must respond exclusively to the measured parameter and not to spurious, high-frequency inputs such as ambient vibration or residual stresses caused by rolling or stretching. If the raw, noisy strain signal is fed directly into a hydraulic actuator, the high-frequency electrical noise translates into physical micro-oscillations at the actuator level. Over time, these localized stresses exacerbate mechanical hysteresis (the non-coincidence of loading and unloading curves), physically degrading the elasticity and accuracy of the material.

Similarly, in advanced semiconductor manufacturing systems utilizing magnetically-levitated linear stages driven by linear hysteresis motors, sub-millimeter precision is required. If the control loop's low-pass filter features a static time constant that is too long, the system suffers from phase delay. In control theory, phase delay catastrophically erodes the phase margin, causing the physical actuator to overshoot its target, overcorrect, and induce a state of continuous, unrecoverable closed-loop oscillation.

To reconcile this, control engineers implement continuous-phase controllers with adaptive filtering. In the design of powered prosthetic knee-ankle legs, first-order Butterworth low-pass filters (typically operating at an 8-Hz cutoff frequency) are used to numerically compute joint velocities. By modulating the smoothing factor dynamically, the prosthetic joint acts essentially rigid and perfectly stable when the user is standing still (filtering micro-tremors to prevent balance loss), yet reacts with near-zero phase latency the moment a forward gait is initiated, utilizing extremum seeking controllers to constantly optimize feedback gains based on the user's immediate comfort and trajectory.

Furthermore, in software engineering, systems like the SWiFT toolkit for operating systems leverage feedback control theory to build adaptive software capable of allocating CPU resources in chaotic shared environments. By incorporating mathematical hysteresis directly into the software controls, TCP congestion algorithms and multi-level feedback schedulers can operate modularly, adapting to drastic changes in network environments (e.g., switching to wireless links) without the unpredictable brittleness associated with static, ad-hoc parameter tuning.

## **Advanced Modalities: Electronic Noses and Fractional-Order Systems**

The application of the Exponential Moving Average and its parameter optimization extends into highly novel sensory and computing domains, showcasing the universal mathematical utility of adaptive baseline tracking.

In environmental recognition, conductimetric electronic noses utilize arrays of sensors to detect volatile organic compounds. However, the baseline resistance of these sensors drifts over time due to thermal changes and prolonged chemical exposure. To extract meaningful features for machine learning classifiers, researchers discretize the time sequences and declare a floating reference resistance. Instead of relying on a static historical baseline, advanced systems utilize the EMA of the resistance as a dynamically floating reference. This allows the system to continuously adapt to gradual steady-state drift (low  $v$ ) while rapidly isolating the sharp relative resistance modulation  $(R(t_0+t) - R_0)/R_0$  triggered by exposure to class-specific target environments.

Furthermore, the boundaries of classical integer-order differential filtering are being expanded by fractional-order systems in the realm of neuromorphic computing. Fractional-order dynamics introduce non-Markovian behavior and scale-free memory into dynamical systems. When integrated into Reservoir Computing (RC) frameworks, fractional-order neurons allow the dynamics of the reservoir to be continuously and systematically tuned via the fractional order  $\alpha$ . By varying this parameter, engineers can uniquely reshape the reservoir's information-processing regime, allowing distinct variables such as memory capacity, active information storage, and information transfer to peak at highly specific points across the operational spectrum.

## Statistical Prediction Regions and Online Parameter Updating

Ultimately, the optimization of a smoothing factor is deeply intertwined with the statistical concept of uncertainty quantification. In machine learning and dynamic time-series forecasting, engineers do not simply generate a point estimate (a single filtered line); they generate prediction regions, mathematically denoted as  $\mathcal{C}_t(X_t; \theta_t, q_\delta, \mathcal{D}_t) \subseteq \mathcal{Y}$ .

In the framework of conformal prediction, the objective is to dynamically construct a boundary set where the true outcome  $Y_t$  will fall within this region with a guaranteed probability of  $1 - \alpha$ , where  $\alpha$  in this specific statistical context represents the target miscoverage rate (and  $\delta a = 1 - \alpha$  is the coverage rate). This is achieved using a non-conformity score function  $s(X_t, y; \theta_t)$ , where the model's internal parameters  $\theta_t$  (which can include the adaptive smoothing factors governing the model's feature inputs) are updated online in real-time. The variable  $q_\delta, \mathcal{D}_t$  represents the  $\delta$  quantile of the dataset. By continually updating the adaptive smoothing factors governing the input data streams, the prediction regions remain tightly bounded during periods of low volatility (minimizing the volume of the set while maintaining the  $1 - \alpha$  coverage rate) and rapidly expand during periods of structural non-stationarity to ensure the true outcome is not missed.

## Conclusion

The mathematical optimization of the smoothing factor  $\gamma$  is a multidimensional problem governed by the inherent physical and statistical tradeoff between system responsiveness and boundary stability. Based on an exhaustive analysis of digital signal processing theory, algorithmic models, metrology, and control system architectures, the conclusions are strictly

defined:

1. **Static Suboptimality and the Variance-Bias Trade-Off:** While a theoretically optimal static smoothing factor can be derived by minimizing a composite risk function containing both steady-state output variance (noise transmission) and dynamic tracking phase delay (lag), this mathematical optimum is strictly dependent on the underlying signal's rate of change. Because real-world environments are inherently non-stationary, any fixed static  $\gamma$  represents a mathematical compromise that is suboptimal across varying operational regimes.
2. **The Superiority of Adaptive Modulation (KAMA):** The true optimum is achieved not through a single static value, but through the deployment of dynamic topologies that calculate  $\gamma_t$  in real-time. In econometrics, Kaufman's Adaptive Moving Average successfully solves this by analyzing fractal path efficiency (the Efficiency Ratio) to dictate an interpolation constant, which is subsequently squared to enforce aggressive boundary stability during noise and rapid tracking during trend breakouts.
3. **Derivative-Driven Cutoff Frequencies (1-Euro):** In hardware kinematics and spatial tracking domains, the 1-Euro Filter presents a rigorous solution by dynamically scaling the filter's continuous-time cutoff frequency based on the smoothed discrete derivative of the input. This mathematically isolates lag reduction and jitter attenuation into decoupled parameters, ensuring precise responsiveness at high velocities and ironclad stability at low velocities.
4. **Physical System and Hardware Implications:** In physical control topologies, the improper selection of  $\gamma$  results in severe physical penalties. In PV+BESS power grids, low smoothing factors induce massive charging currents that saturate battery state-of-charge, causing grid compliance failure. In electromechanical systems driven by hysteresis motors or continuous-phase prosthetic joints, static phase-delays induce catastrophic closed-loop instability and physical hardware wear. Adaptive smoothing is a mathematically mandatory prerequisite stage for robust controllers like Model Predictive Control (MPC) and Active Disturbance Rejection Control (ADRC).
5. **Computational Implementation Optimization:** When mathematical theory is deployed to hardware, physical constraints dictate the software architecture. Approximating the smoothing factor to base-2 inverse exponential parameters ( $\gamma = 1/2^k$ ) allows the implementation of integer bit-shifting algorithms that bypass floating-point arithmetic bottlenecks. This preserves critical computational bandwidth in high-frequency logic controllers executing flowmeter conversions or active disturbance compensation routines.

Ultimately, the optimal  $\gamma$  is not a fixed integer or fraction, but rather the continuous execution of an algorithmic architecture capable of calculating its own parameters dynamically. Only by directly incorporating mathematical feedback regarding the signal's instantaneous volatility, fractal efficiency, and velocity can a filter simultaneously satisfy the conflicting physics of boundary stability and real-time responsiveness.

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